GENERAL CHEMISTRY CONSOLIDATED REVISION CHEAT SHEET

Grounded strictly in uploaded course sources: Chapters 2, 3, 4, 10, 11, 13, 15, and 16

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1. All Formulas and Equations

Topic / Chapter	Formula / Equation	Variables, Meaning & Units
Atomic Structure (Ch 2)	$A = Z + N$	A: Mass number, Z: Atomic number (protons), N: Neutrons.
Atomic Mass Unit (Ch 2)	$1\ \text{amu} = \frac{1}{12} \times m(^{12}\text{C}) = 1.66054 \times 10^{-24}\ \text{g}$	Standard atomic mass unit based on Carbon-12.
Molecular Weight (Ch 2)	$\text{MW} = \sum(\text{Atomic Weight} \times \text{Count})$	Sum of atomic masses for all atoms in molecular formula.
Moles & Mass (Ch 3)	$n = \frac{m}{M}$	n: Moles (mol), m: Mass (g), M: Molar mass (g/mol).
Percent Yield (Ch 3)	$\%\ \text{Yield} = \left(\frac{\text{Actual Yield}}{\text{Theoretical Yield}}\right) \times 100\%$	Actual vs. calculated maximum yield (same mass units).
Molarity (Ch 4)	$M = \frac{n}{V}$	M: Molarity (mol/L or M), n: Solute moles, V: Volume (L).
Dilution Equation (Ch 4)	$M_i V_i = M_f V_f$	M_i, V_i: Initial molarity & volume; M_f, V_f: Final molarity & volume.
Pressure Definition (Ch 10)	$P = \frac{F}{A}$	P: Pressure (Pa or N/m^2), F: Force (N), A: Area (m^2).
Ideal Gas Law (Ch 10)	$PV = nRT$	P: atm, V: L, n: mol, $R = 0.08206\ \text{L} \cdot \text{atm}/(\text{mol} \cdot \text{K})$, T: K.
Gas Density (Ch 10)	$d = \frac{PM}{RT}$	d: Density (g/L), M: Molar mass (g/mol).
Molar Mass from Gas (Ch 10)	$M = \frac{mRT}{PV} = \frac{dRT}{P}$	Calculated molecular weight from gas state variables.
Mass Percentage (Ch 13)	$\text{Mass}\ \% = \frac{\text{Mass of Solute}}{\text{Total Solution Mass}} \times 100\%$	Mass fraction multiplied by 100.
Trace Concentration (Ch 13)	$\text{ppm} = \frac{\text{Solute Mass}}{\text{Solution Mass}} \times 10^6$	Parts per million (10^9 for ppb).
Equilibrium Constant (Ch 15)	$K_c = \frac{[\text{C}]^c[\text{D}]^d}{[\text{A}]^a[\text{B}]^b}$	Reaction $a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$. Pure (s) and (l) omitted.
pH & pOH Scale (Ch 16)	$\text{pH} = -\log[\text{H}_3\text{O}^+], \quad \text{pOH} = -\log[\text{OH}^-]$	Logarithmic measure of hydronium and hydroxide molarity.
pH-pOH Relation (Ch 16)	$\text{pH} + \text{pOH} = 14.00 \quad (25^\circ\text{C})$	Derived from $K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14}$.

2. All Key Definitions

- **Law of Conservation of Mass (Ch 2)**: Total mass of substances present at the end of a chemical process equals the total mass present before the reaction.
- **Law of Definite / Constant Composition (Ch 2)**: A given chemical compound always contains the exact same elements in the exact same mass proportions.
- **Law of Multiple Proportions (Ch 2)**: When two elements combine to form more than one compound, the masses of one element combining with a fixed mass of the other are in small whole-number ratios.
- **Isotopes (Ch 2)**: Atoms of the same element (Z) containing different numbers of neutrons in their nuclei (different mass numbers A).
- **Combination / Decomposition / Combustion Reactions (Ch 3)**: Combination forms 1 product ($A + B \rightarrow C$); Decomposition breaks 1 substance into multiple products ($C \rightarrow A + B$); Combustion reacts hydrocarbons with O_2 producing CO_2 and H_2O.
- **Mole & Avogadro's Number (Ch 3)**: The amount of matter containing 6.022×10^{23} structural units (atoms, molecules, or ions).
- **Limiting Reactant vs. Excess Reagent (Ch 3)**: Limiting reactant is completely consumed and limits product quantity; excess reagent remains partially unreacted.

- **Theoretical vs. Actual Yield (Ch 3)**: Theoretical yield is the calculated maximum product mass based on stoichiometry; actual yield is the measured laboratory amount.
- **Solution, Solute, Solvent (Ch 4, 13)**: Solution is a homogeneous mixture; solute is the component in smaller amount; solvent is the component in larger amount (water in aqueous solutions).
- **Electrolyte vs. Nonelectrolyte (Ch 4)**: Electrolyte yields ions in aqueous solution and conducts electricity; nonelectrolyte dissolves without producing ions.
- **Oxidation vs. Reduction (Ch 4)**: Oxidation is electron loss (increase in oxidation state); reduction is electron gain (decrease in oxidation state).
- **Effusion vs. Diffusion (Ch 10)**: Effusion is the escape of gas molecules through a microscopic orifice into a vacuum; diffusion is the movement/spreading of gas particles through space.
- **Intermolecular Forces (Ch 11)**: Attractive forces between neighboring molecules (dispersion, dipole-dipole, hydrogen bonding) controlling physical states and boiling points.
- **Chemical Equilibrium (Ch 15)**: Dynamic state where forward and reverse reaction rates are equal, maintaining constant reactant and product concentrations.
- **Amphiprotic Species (Ch 16)**: A molecule or ion capable of acting as either a proton donor (acid) or a proton acceptor (base) (e.g., H_2O, HCO_3^-).
- **Conjugate Acid-Base Pair (Ch 16)**: Two chemical species that differ only by the presence or absence of a single proton (H^+).

3. All Laws, Principles, and Rules

1. **Dalton's Atomic Postulates (Ch 2)**: Matter consists of indivisible atoms. All atoms of an element are identical. Atoms combine in fixed whole-number ratios to form compounds. Atoms are rearranged (not created/destroyed) in reactions.
2. **Gas Laws (Ch 10)**:
   - **Boyle's Law**: $V \propto 1/P$ (constant T, n).
   - **Charles's Law**: $V \propto T$ (constant P, n).
   - **Avogadro's Law**: $V \propto n$ (constant P, T).
3. **Le Châtelier's Principle (Ch 15)**: If a stress (change in temperature, pressure, or concentration) is applied to a system at equilibrium, the system shifts its position to counteract the stress.
4. **Oxidation State Rules (Ch 4)**:
   - Elementary forms (O_2, Fe) have oxidation number 0.
   - Monatomic ions equal their ionic charge.
   - Oxygen is -2 in most compounds (exceptions: -1 in peroxides H_2O_2, $-1/2$ in superoxides KO_2).
   - Hydrogen is $+1$ with nonmetals, -1 with metals.
   - Fluorine is always -1.
   - Neutral molecules sum to 0; polyatomic ions sum to net charge.

4. Main Headings and Key Concepts by Chapter

- **Chapter 2 (Atoms, Molecules, Ions)**: Subatomic particles (p^+, n^0, e^-); atomic symbol notation ${}^A_Z\mathrm{X}$; isotopic abundance; empirical vs. molecular formulas.
- **Chapter 3 (Stoichiometry)**: Balancing equations; mole conversions ($n = m/M$); percent composition; empirical formula determination; limiting reagent calculations; percent yield.
- **Chapter 4 (Aqueous Solutions)**: Homogeneous solutions vs. suspensions; strong vs. weak electrolytes; precipitation reactions; oxidation-reduction balancing; titration stoichiometry.
- **Chapter 10 (Gases)**: Gas properties; barometer/manometer pressure units; $PV = nRT$; gas density $d = PM/(RT)$; Dalton's law of partial pressures; effusion rates.
- **Chapter 11 (Liquids & Intermolecular Forces)**: Condensed phases; dispersion, dipole-dipole, and hydrogen bonding; dynamic vapor pressure; boiling point definitions; phase diagrams.
- **Chapter 13 (Properties of Solutions)**: "Like dissolves like"; solvation energetics; temperature effects on solubility (solids increase, gases decrease); mass %, ppm, ppb.
- **Chapter 15 (Chemical Equilibrium)**: Reversible reactions; equilibrium constant expression K_c; reaction quotient Q; heterogeneous equilibria; Le Châtelier shifts.
- **Chapter 16 (Acid-Base Equilibria)**: Arrhenius, Brønsted-Lowry, and Lewis acid-base definitions; autoionization of water (K_w); pH and pOH calculations; conjugate pairs.

5. Units, Constants, and Conversion Factors

Fundamental Constants and Unit Equivalencies

- **Avogadro's Number (N_A)**: 6.022×10^{23} particles/mol
- **Atomic Mass Unit (1 amu)**: 1.66054×10^{-24} g
- **Proton Mass**: 1.00728 amu = 1.6726×10^{-24} g
- **Neutron Mass**: 1.00867 amu = 1.6749×10^{-24} g
- **Electron Charge**: -1.602×10^{-19} C
- **Gas Constant (R)**: $0.08206\ \text{L} \cdot \text{atm}/(\text{mol} \cdot \text{K})$
- **Water Ion Product (K_w)**: 1.0×10^{-14} at 25°C
- **Pressure Equivalencies**:
  1.00 atm = 760 torr = 760 mmHg = 101.325 kPa = 1.01325 bar
  $1\ \text{Pa} = 1\ \text{N/m}^2, \quad 1\ \text{bar} = 10^5\ \text{Pa}$
- **Temperature**: $T(\text{K}) = t(^\circ\text{C}) + 273.15$
- **STP**: 0°C (273.15 K) and 1.00 atm (760 torr)

6. Classifications, Conditions, and Exceptions

- **Electrolyte Classifications**:
  - *Strong Electrolytes*: Completely ionized in water ($HCl, HNO_3, HClO_4, H_2SO_4$, Group 1A/2A hydroxides, soluble salts).
  - *Weak Electrolytes*: Partially ionized ($HF, CH_3COOH, H_2SO_3, NH_3$).
  - *Nonelectrolytes*: Dissolve as neutral molecules without ionization (C_2H_5OH, sucrose).
- **Heterogeneous Equilibrium Exclusions**: Pure solids (s) and pure liquids (l) have fixed intensive concentrations and are **excluded** from K_c and K_p expressions.
- **Catalysts in Equilibrium**: Catalysts increase forward and reverse rates equally; they shorten the time to reach equilibrium without altering K_c or equilibrium position.
- **Gas Solubility Exception**: Unlike solid solutes (whose solubility usually increases with temperature), gas solubility **decreases** as temperature rises.

7. Important Concept Comparisons

Table 7.1: Comparison of Acid-Base Models (Ch 16)

Model	Acid Definition	Base Definition
Arrhenius	Produces H^+ in water	Produces OH^- in w
Brønsted–Lowry	Proton (H^+) donor	Proton (H^+) accept
Lewis	Electron-pair acceptor	Electron-pair donor

Table 7.2: Subatomic Particles Summary (Ch 2)

Particle	Location	Mass (amu)	Mass (g)	Relative
Proton (p^+)	Nucleus	1.00728	1.6726×10^{-24}	1837 × el
Neutron (n^0)	Nucleus	1.00867	1.6749×10^{-24}	1842 × el
Electron (e^-)	Extra-nuclear	0.00055	9.109×10^{-28}	1 × (Base